

Multiple Choice (10 marks)

1. What factors affect the rate of transpiration?
 - (a) Temperature, humidity, and air currents
 - (b) Plant species, root depth, and wind speed
 - (c) Soil fertility, irrigation methods, and cloud cover
 - (d) Water salinity, leaf color, and time of day
 - (e) Leaf surface area, stem length, and moon phase
2. What are the different types of point mutations?
 - (a) Frame shift mutations and nucleotide repeats
 - (b) Base substitutions and small insertions and deletions
 - (c) Base additions and large insertions
 - (d) Silent mutations and missense mutations
 - (e) Duplication mutations and triplet repeat expansions
3. Which is a true statement concerning genetic variation?
 - (a) It is created by the direct action of natural selection.
 - (b) It tends to be reduced by the processes involved when diploid organisms produce gametes.
 - (c) It must be present in a population before natural selection can act upon the population.
 - (d) It is always increased by the processes involved when diploid organisms produce gametes.
 - (e) It is not affected by the environmental changes.
4. Which group is composed entirely of individuals who maintained that species are fixed (i.e., unchanging)?
 - (a) Aristotle, Darwin, and Lamarck
 - (b) Aristotle, Lyell, and Darwin
 - (c) Aristotle, Linnaeus, and Cuvier
 - (d) Aristotle, Cuvier, and Lamarck
 - (e) Linnaeus, Darwin, and Lamarck
5. Compare the methods of obtaining oxygen in the earthworm, the frog, the fish and man.

- (a) Earthworms use gills, frogs use both gills and lungs, fish use lungs, and humans use skin.
- (b) Earthworms and humans use lungs, frogs and fish use gills.
- (c) All use gills for obtaining oxygen.
- (d) Earthworms and humans use gills, frogs use skin, and fish use lungs.
- (e) Earthworms use lungs, frogs use gills, fish use skin, and humans use gills.

True / False (10 marks)

Answer True or False

- 6. True or False: Search engine as a diagnostic tool in difficult immunological and allergologic cases: is Google useful?
- 7. True or False: Can increases in the cigarette tax rate be linked to cigarette retail prices?
- 8. True or False: Are head and neck specific quality of life measures necessary?
- 9. True or False: Did Chile's traffic law reform push police enforcement?
- 10. True or False: Is there a correlation between androgens and sexual desire in women?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the role of radioactive isotopes in biological research, particularly in tracing compound pathways in living cells, and analyze their impact on cell observation techniques.
- 12. Discuss the role of ethylene in the ripening process of fruits and evaluate the misconceptions surrounding the use of heat from kerosene stoves in this context.

End of Paper